DS 6050: Deep Learning

School of Data Science, University of Virginia

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- **Asynchronous Lectures & Readings:** Available weekly on Canvas.
- **Course Website:** Link

Other Resources:
- Assignments submission (UVA Canvas)

Why Should You Care About Learning Deep Learning?

How can you design an AI that truly understands complex medical images, not just mimics patterns? How do you build a language model that can generate coherent text, or an algorithm that can innovate in scientific discovery? These are the kinds of "grand challenges" that today's data scientists and ML engineers tackle, requiring more than just off-the-shelf code. They demand a deep, quantitative understanding of how these models work, from first principles.

You've likely seen countless AI tutorials online, much like the millions of fitness videos on YouTube. Yet, professional athletes and Olympic medalists dedicate themselves to rigorous training camps with expert coaches. Why? Because true mastery, the kind that forges resilient skill and deep intuition, isn't about mimicking surface-level actions. It's built on structured guidance, disciplined hands-on practice with expert feedback, and a grasp of fundamentals.

This course is your intensive "training camp" for deep learning. It's where you'll move beyond superficial understanding to forge genuine expertise, from the core out, embracing the deliberate practice needed to push your boundaries and achieve lasting skill.

How Will This Course Help You Succeed (As a Data Scientist/ML Engineer)?

Grand challenges in AI require data scientists who can think critically, design robust systems, and make informed trade-offs. This course will equip you with a conceptual and practical framework to tackle such complex problems in your future research, engineering practice, or advanced studies.

By the end of this training camp, you will be better able to answer critical questions like:

1. How do I use math and core principles to truly understand why, how, and where deep learning models succeed or fail? (No more magical black boxes!)
2. How can I build fundamental components (like backpropagation or an attention mechanism) "from scratch" to solidify my understanding?
3. Beyond the buzzwords and LinkedIn cheatsheets, how do I develop the critical judgment to make sound engineering decisions, evaluate new research, and maintain my intellectual integrity?
4. How do I use the foundational skills and "learning to learn" strategies from this course to confidently tackle future AI advancements and projects long after graduation?

Course Structure & How You'll Master Deep Learning

This course follows a carefully designed 12-module progression that builds deep learning expertise through three integrated phases:

Phase 1: Foundations & From-Scratch Understanding (Modules 1-3) Master the mathematical underpinnings and implement core algorithms from scratch. You'll build linear models, multi-layer perceptrons, and backpropagation using only NumPy, developing deep intuition about how and why deep learning works. Critical emphasis on optimization foundations and ablation methodology that will recur throughout the course.

Phase 2: Architectural Innovations & Domain Specialization (Modules 4-9) Explore how different data modalities drive architectural choices. From CNNs for spatial data to RNNs for sequences to Transformers for universal modeling, you'll understand why specific architectures excel at specific tasks. Each module revisits optimization challenges unique to that architecture, building your ability to diagnose and fix real-world training problems.

Phase 3: Modern Practice & Research Skills (Modules 10-12) Master contemporary techniques including vision transformers, large-scale pretraining, and generative models. Learn systematic ablation studies, paper reproduction, and research methodology. The course culminates with your team tackling a "grand challenge" using all the skills you've developed.

Course Calendar

Table 1: Course Schedule

Week	Module	Topic
Week 1	Module 1	From Linear Regression to Neural Networks
Week 2	Module 2	Backpropagation
Week 3	Module 3	Training MLPs
Week 4	Module 4	Convolutional Neural Networks
Week 5	Module 5	Advanced CNN Architectures
Week 6	Module 6	Encoder Decoder Architectures
Week 7	Module 7	Recurrent Neural Networks
Week 8	Module 8	Attention Mechanism
Week 9	Module 9	Self-Attention and Transformers
Week 10	Module 10	Transformer Models
Week 11	Module 11	Prompting, PEFT & Quantization
Week 12	Module 12	Generative Modeling

Assessment as Learning: How You'll Build and Demonstrate Mastery

Assessment in this course is designed primarily as a learning tool, with evaluation as a by-product. Each assessment provides opportunities to deepen understanding, receive feedback, and refine your skills.

Programming Assignments: Deliberate Practice with Feedback

Five comprehensive assignments that build your implementation skills progressively:

- **Learning Focus:** Each assignment includes extensive unit tests that guide your implementation and provide immediate feedback
- **Iterative Development:** You can discuss together and explore ideas in the forum
- **Reflection Component:** Brief write-ups explaining your design choices and debugging process
- **Peer Code Review:** Optional exchanges with classmates to learn from different approaches

Assignment Progression (current format):

1. Foundations & Backprop (Modules 1–2) → Homework 1 (Colab on Module 1 page): linear models, gradients, and early autograd practice.
2. Optimization & CNNs (Modules 4–5) → Homework 2 (Colab on Module 5 page): training stability, convolutional stacks, and transfer learning.
3. Seq2Seq (Module 7) → Homework 3: seq2seq (Colab). Link: Colab notebook.
4. Attention (Modules 8) → Homework 4: Add cross-attention to your GRU-based seq2seq model
5. Module 10 → Homework 5: Transformer 2.0 (Colab). Link: Colab notebook.

Module Quizzes: Knowledge Reinforcement

- **Purpose:** Reinforce key concepts from asynchronous content and readings
- **Format:** Short (10-15 questions), open-book, focused on understanding not memorization
- **Completion-Based:** Full credit for thoughtful completion, encouraging exploration over perfection
- **Immediate Feedback:** Explanations provided for all answers to support learning

Group Project: Solving a Grand Challenge

Multi-phase project applying course concepts to real-world problems:

Formative Milestones:

1. **Proposal & Literature Review:** Receive feedback on problem framing and approach
2. **Mid-Project Check-in:** Share challenges and get guidance from peers and instructors
3. **Final Deliverables:** Complete solution with documentation and reflection

Grading: How Your Learning Translates to Grades

While assessment focuses on learning, grades provide a summary of your demonstrated mastery. The grading schema and Late Policy and Extensions is the standard MSDS practice. Here is the breakdown for the total grade:

Component	**Weight**	**Details**
Programming Assignments	40%	Sum of grades
Group Project	40%	Proposal (5%), Mid-checkpoint (10%), Final deliverables (25%)
Participation	20%	Quiz + engagement.

Professional and Academic Integrity

As future leaders in data science and AI, uphold the highest standards of ethics and integrity. All work must comply with the UVA Honor System. Individual assignments must be your own original work, though conceptual discussions are encouraged. Any external code, ideas, or resources must be appropriately cited. The pledge should be included with relevant submissions.